

Volatility Forecasting: A Statistical and Deep Learning Approach

Abstract

1 Introduction

1.1 Context

Modern share trading is often traced to Amsterdam in 1602, when the Dutch East India Company issued transferable shares and investors began trading them with one another [1]. At that point in time, it originated the possibility to earn money by buying the paper at a lower price, hoping that the company would grow with its expansion. Or, if someone had privileged information, e.g. a failed expedition or a successful one, would be in a better position than the others, being able to sell its position before it loses value or buy more before it gets a higher price.

The desire to obtain and interpret information that may affect stock prices remains central to financial markets. Services such as the Bloomberg Terminal give investors rapid access to news and data [2], while some traders use direct exchange feeds when even small delays matter [3]. If you receive the information before every player, you have a leverage over them and can position yourself in a beneficial way. Jim Simons, the founder of Renaissance Technologies, whose fund Medallion had an average annual return of 66% before investor fees, started his trading using news in early human trading decisions, before passing to models who studies historical market data as stated by Gregory Zuckerman in his Jim Simons biography, *The Man Who Solved the Market* [4].

As competition for information increases, the opportunity to profit from that same information decreases. Fama's efficient-market hypothesis states that prices reflect available information; in its weak form, this includes information in past prices [5]. An efficient market is one where prices adjust

quickly enough to available information such that an investor cannot reliably earn extra returns from it after accounting for risk and trading costs. Historical prices can still be studied for candidate patterns, but market efficiency gives no reason to assume that those patterns will yield profits. As described by Zuckerman, Simons searched historical price data for recurring patterns [4].

A historical data is known as a time series. Time series analysis has been a widely explored field in statistics and ML (Machine Learning). Statistical analysis is focused on understanding the properties of the data (stationarity, mean reversion, seasonality, trend, cycles, etc.) and creating a model, based on such characteristics that successfully explains the movements observed in such data. ML, in the meanwhile, is interested in creating a model that efficiently explains the data without effectively needing its traits.

1.1.1 Stastical Models

An early landmark of time series statistical modelling is Yule’s 1927 autoregressive analysis, where the current value depends on past values [6]. In 1927, Slutsky demonstrated that summing random values can produce smooth, apparently cyclical movements [7], and in 1938, Wold showed that a weakly stationary time series can be split into a predictable component and another represented by current and past shocks [8]. Weak stationarity means a constant mean and finite variance, with covariance depending only on time lag. Both contributions served as foundations for moving-average representations.

Box and Jenkins’s 1970 treatment popularized ARIMA (Autoregressive Integrated Moving Average) modeling, combining AR and MA terms with differencing [9]. In ARIMA(p, d, q), p and q correspond to the orders of AR and MA, respectively, while d is the number of differences used to address some forms of non-stationarity, especially stochastic trends. Their framework also treats seasonal ARIMA models. SARIMA(p, d, q)(P, D, Q)_s uses (p, d, q) for the nonseasonal component. Seasonality is a pattern that tends to recur every s observations; for monthly data with a yearly pattern, s = 12. P and Q are the orders of seasonal AR and MA parts, respectively, and D is the number of seasonal differences.

Box and Jenkins’s models address the series’ expected value over time. In a homoscedastic specification, the unexplained component is modeled as white noise ($\varepsilon_t \sim \text{WN}(0, \sigma^2)$) with constant variance. Robert Engle’s 1982 ARCH (Autoregressive Conditional Heteroskedasticity) model instead studies time-varying conditional variance, originally in the context of uncertainty about future U.K. inflation [10]. A shock can have constant unconditional

variance while its variance conditional on past information changes over time. ARCH captures this by relating current conditional variance to past squared shocks.

Volatility often stays elevated for many periods, so ARCH can require many past squared shocks and consequently many parameters. Volatility is the standard deviation of returns, equal to the square root of their variance. Tim Bollerslev’s 1986 GARCH (Generalized Autoregressive Conditional Heteroskedasticity) model added lagged conditional variance terms to represent this persistence parsimoniously [11].

In 2009, Fulvio Corsi presented the HAR (Heterogeneous Autoregressive) model. Market participants act over different time horizons, so realized volatility measured over a day, a week, and a month may help predict future realized volatility [12]. Corsi drew on the heterogeneous-market hypothesis discussed by Ulrich A. Müller et al. in their 1993 working paper *Fractals and Intrinsic Time: A Challenge to Econometricians* [13].

AR, MA, ARIMA and SARIMA models the conditional mean. HAR presents a Realized-volatility regression. ARCH and GARCH the conditional variance. All shocks were modeled as WN. However, there are models who assume that shocks behaves like a Brownian Motion, mostly in econometry.

In 1900, Bachelier modeled speculative price changes using a Brownian-motion precursor [14]. In 1959, Osborne studied Brownian behavior in log price changes [15]. Samuelson’s 1965 warrant-pricing model used lognormal asset prices, corresponding to geometric Brownian motion [16]. In 1973, Fischer Black and Myron Scholes derived the theoretical price of a European call option on a non-dividend-paying stock under a geometric-Brownian-motion price model [17]. In 1998, Fabienne Comte and Eric Renault presented a continuous-time stochastic-volatility model with long memory [18]. Gatheral, Jaisson, and Rosenbaum’s 2014 preprint, published in 2018, proposed rough fractional stochastic volatility with a Hurst exponent below one half [19].

1.1.2 ML Models

In Machine Learning, the main idea is to create a model able to learn patterns from data. Neural networks combine weighted inputs through nonlinear layers, and their parameters can be adjusted by backpropagation. Werbos described the method in his 1974 dissertation [22]; Rumelhart, Hinton, and Williams demonstrated its use for learning internal representations in multi-layer networks in 1986 [23].

In 1987, Lapedes and Farber studied feedforward neural networks for nonlinear signal prediction and system modeling [24]. Waibel et al. used a time-delay neural network (TDNN) for phoneme recognition in 1989, apply-

ing shared computations across successive time positions [25]. Bai, Kolter, and Koltun evaluated a generic temporal convolutional network (TCN) with causal, dilated convolutions in 2018 [26]. Reisenhofer et al. developed HAR-Net in 2022, using causal convolutional layers that can be initialized to reproduce a fitted HAR forecast before optimization [27].

Elman’s 1990 recurrent neural network illustrates how a model can process sequential observations while carrying a hidden state forward [28]. Standard RNNs can have difficulty learning long-range dependencies because gradients may vanish or explode. Hochreiter and Schmidhuber introduced the LSTM (Long Short-Term Memory) in 1997 to help preserve error signals over longer spans [29]. Cho et al. introduced the gated recurrent unit (GRU) in 2014, using a simpler gated state than the LSTM; its parameter count and training speed depend on the chosen architecture and implementation [30].

In 2001, Breiman presented random forests, which aggregate predictions from randomized decision trees; for regression, their outputs are averaged [31]. Chen and Guestrin introduced XGBoost in 2016 as a scalable implementation of gradient-boosted trees, fitted sequentially to improve an objective function [32].

Vaswani et al. introduced the Transformer in 2017. Its self-attention relates positions in a sequence without a recurrent pass; autoregressive forecasting additionally requires a causal attention mask [33].

Each of these models have been used to predict returns or volatility even if they are not explicitly built for that reason as a HAR model is, for example, and that’s the main reason they are useful. They are not limited to one scope. One of the main benefits of ML models is generality.

1.1.3 Estimating Volatility

Unlike prices and returns, volatility is latent. It describes the dispersion of possible returns, so a single observed return cannot reveal its value: a single return reflects both the prevailing volatility and the particular random shock that occurred. In a continuous-time model, instantaneous volatility may also change within a trading day. Consequently, neither a daily closing price nor one daily return identifies the variance accumulated over that day.

Daily variance must therefore be estimated from observable prices. With sufficiently frequent observations and under suitable assumptions, the sum of squared intraday log returns approximates the day’s integrated variance; this is known as realized variance [34]. When intraday data are unavailable, range-based estimators use information in daily prices: Parkinson uses the high and low [35], while Garman–Klass uses the open, high, low, and close [36]. All are estimates rather than exact observations, and their accuracy

depends on the available data and the assumptions behind each estimator.

1.2 Motivations

The main objective with this project is understanding the better approach for predicting the price of an asset. The potential capabilities of ML models in finance forecasting tasks are tremendous, although the stability and robustness of statistical models are extremely reliable.

Several approaches have been presented as comparing both types of models. However, our objective is to extend these comparisons to include, besides the baseline models (AR, SARIMA, GARCH, HAR, MLP), HARnet and RFSV models.

Furthermore, we are interested in understanding the generalization capacity of such models. Our focus is to understand the difference in performance between a local (fitted by one equity data) and a global (fitted by several tickers) model.

1.3 Contributions

In this paper, we present the comparison between AR(1), SARIMA(1,0,1)(1,0,1,5), GARCH(1,1), HAR-20 (base HAR using averages from 1, 5 and 20 observations), HAR-80 (HAR-20 extended to 40 and 80 observation averages), RFSV, LSTM, SiLU-LSTM (tanh activation function changed by SiLU), HARnet-20, HARnet-80. Each model will be fitted locally, one model fitted per each of 5 top-volume tickers, and globally, across a dataset of 2.714 tickers from NYSE and NASDAQ currency exchanges.

1.4 Related Work

Corsi’s heterogeneous autoregressive (HAR) model uses averages of realized volatility over daily, weekly, and monthly horizons to approximate persistent volatility dynamics with a small number of coefficients. Its empirical fits use intraday data for the USD/CHF spot exchange rate (Reuters quotes, 1989–2003), S&P 500 index futures (1990–2007), and 30-year U.S. Treasury bond futures (1990–2003); the equity exposure is an index future, not individual stocks [12]. Audrino and Chassot compare HAR with machine-learning methods using five-minute returns from NYSE TAQ trades in 2015–2023 for 1,445 U.S. stocks matched to CRSP. Their full sample includes 27 Dow Jones Industrial Average constituents and 69 Nasdaq-100 constituents, but is not restricted to those indexes. They show that the HAR fitting window

and re-estimation frequency materially affect the comparison; in their setting, a carefully fitted HAR remains a strong benchmark [37]. Reisenhofer, Bayer, and Hautsch extend this baseline with HARNet, a hierarchy of dilated causal convolutions initialized to reproduce a fitted HAR forecast before further optimization. They fit and test both 20- and 80-observation variants on five-minute realized-variance series for the S&P 500, FTSE, and Dow Jones Industrial Average indexes from the Oxford-Man Institute’s Realized Library, covering 2001–2020. These are index series rather than individual equities [27].

The choice between fitting one model per asset and pooling assets is another central question. Montero-Manso and Hyndman define a *local* method as fitting a separate forecasting function to each series and a *global* method as fitting one function using observations pooled across series. They show that a sufficiently flexible global method can represent the same forecasts as local methods even when the series differ. With a fixed finite input window, however, one global function must give the same forecast to identical inputs, whereas two local functions can differ; longer histories or series identifiers can distinguish such cases. Pooling increases the number of training examples available for shared parameters, but this is a modeling opportunity, not a guarantee that a global fit will forecast better. Their experiments use non-equity benchmarks including M1–M4 competition series, ATM cash demand, Wikipedia traffic, weather, and electricity demand, so their results motivate the local-versus-global design rather than directly establish its performance for stocks [38]. Liu et al. use 2014–2023 daily closing prices from Reuters Refinitiv Workspace for 11,771 individual stocks from exchanges in Asia, Europe, North America, and elsewhere; 10,240 series are available for global training and 1,531 stocks are reserved for unseen-stock evaluation. They report benefits from larger and more heterogeneous training pools [39]. Zhang et al. use Nasdaq ITCH order-book data supplied by LOBSTER for 93 S&P 500 constituent stocks selected from the 100 most liquid, spanning July 2011 to June 2021. They compare stock-specific and pooled models for intraday realized volatility, add a market-volatility proxy, and test previously unseen stocks [40]. These studies motivate evaluating local and pooled models separately, while recognizing that intraday predictors are not available for every asset in the present equity dataset.

Rough-volatility studies provide a complementary statistical approach. Gatheral, Jaisson, and Rosenbaum estimate roughness from DAX equity-index and German Bund bond futures using QuantHouse intraday data (2010–2014), and from S&P and Nasdaq index realized-variance series in the Oxford-Man library (2000–2014). They find evidence that log-volatility is rough and propose a rough fractional stochastic-volatility model with a

Hurst exponent below one half; this is estimation and model calibration, not neural-network training [19]. Mouti investigates roughness using Bloomberg daily open, high, low, and close data from April 2005 to April 2015. The asset set includes U.S. and international equity indexes (S&P 100, S&P 400, IBEX 35, FTSE 100, Hang Seng, and others) and individual shares such as Google and Microsoft; the main illustrations use S&P 100 and IBEX 35. Mouti also compares rough-volatility forecasts with AR, HAR, and GARCH forecasts [41]. Tang, Rosenbaum, and Zhou compare a pooled LSTM with a parsimonious rough-volatility model using five-minute Binance prices for 213 cryptocurrencies quoted against USDT. Their training period is 2020–2021 and their test period covers the first half of 2022; no equities or equity indexes are in that sample [42].

Finally, Brini compares pretrained, zero-shot time-series foundation models with econometric realized-volatility benchmarks on the VOLARE evaluation set: 40 individual U.S. stocks from all 11 GICS sectors (2015–2026), five major foreign-exchange pairs, and five commodity or equity-index futures. VOLARE’s five-minute realized measures are derived from Kibot tick data. The 50 assets are used for evaluation and for rolling estimation of the econometric benchmarks, not for task-specific training of the zero-shot foundation models. Brini finds that conclusions depend on how errors are aggregated across assets and whether forecasts are recalibrated [43]. The present thesis investigates models fitted on its own daily variance proxies, with local and global equity fits and matched forecast observations as the basis for eventual comparison. No result from the cited papers establishes the ranking of those models on this dataset.

1.5 Outline

On Section 2, we’ll be explaining the concepts behind volatility prediction, starting from statistical theory until ML theory necessary to understand the paper. On Section 3, we’ll be describing each model and our framework for evaluation. On Section 4, we’ll focus on implementation and reproducibility. On Section 5, we present our results. On Section 6, conclusion and further work.

2 Background

2.1 Statistics Concepts

Let $Y(t)$ be a random variable observed over time, and let $\mathcal{F}(t-1)$ contain the information available before time t . An unconditional property describes the distribution of $Y(t)$ without using that history; a conditional property describes its distribution given $\mathcal{F}(t-1)$. For example, $E[Y(t)]$ is an unconditional mean, whereas $E[Y(t) \mid \mathcal{F}(t-1)]$ is the mean predicted from past information. A moment summarizes an aspect of a distribution: its first moment concerns location, its second central moment dispersion, and its third central moment asymmetry.

The population mean is $\mu = E[Y(t)]$, and the sample mean of n observations is $\bar{y} = n^{-1} \sum_{t=1}^n y(t)$. An autoregressive model, for example $Y(t) = c + \phi Y(t-1) + \varepsilon(t)$ with $E[\varepsilon(t) \mid \mathcal{F}(t-1)] = 0$, has conditional mean $c + \phi Y(t-1)$. SARIMA extends autoregressive and moving-average modeling with ordinary and seasonal differencing; it likewise specifies a forecast of the conditional mean of the modeled series. Differencing must be reversed when expressing a forecast on the original scale [9].

The population variance is $\text{Var}(Y(t)) = E[(Y(t) - \mu)^2]$. Its usual unbiased sample estimator for independent, identically distributed observations with finite variance is $s^2 = (n-1)^{-1} \sum_{t=1}^n (y(t) - \bar{y})^2$. For dependent time-series observations, this formula remains a descriptive sample variance, but its unbiasedness is not automatic. Conditional variance describes uncertainty left after observing the past: $h(t) = \text{Var}(Y(t) \mid \mathcal{F}(t-1)) = E[\varepsilon(t)^2 \mid \mathcal{F}(t-1)]$ when $\varepsilon(t) = Y(t) - E[Y(t) \mid \mathcal{F}(t-1)]$. ARCH models $h(t)$ from past squared shocks, for example $h(t) = \omega + \alpha \varepsilon(t-1)^2$; GARCH also uses past conditional variance, for example $h(t) = \omega + \alpha \varepsilon(t-1)^2 + \beta h(t-1)$. With $\omega > 0$, $\alpha, \beta \geq 0$, and $\alpha + \beta < 1$ (taking $\beta = 0$ for ARCH), these models can have constant unconditional variance while conditional uncertainty changes [10, 11].

Covariance generalizes variance to two variables: $\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$, so $\text{Var}(X) = \text{Cov}(X, X)$. Applied to $Y(t)$ and $Y(t-k)$, it measures linear dependence across a lag k . The standardized third central moment, $E[(Y(t) - \mu)^3] / \text{Var}(Y(t))^{3/2}$, is skewness when the variance is positive and the third moment exists. Its sign indicates the direction of asymmetry.

A shock is the part of an observation that the conditional-mean model does not predict. Zero-mean white noise has constant finite unconditional variance and zero covariance between distinct dates. White noise need not be Gaussian or independent. In a homoscedastic SARIMA model, the innovations are commonly modeled as white noise with constant conditional

variance. In ARCH and GARCH, the raw shocks $\varepsilon(t) = \sqrt{h(t)}z(t)$ can be uncorrelated with constant unconditional variance while their conditional variance $h(t)$ changes; the standardized innovations $z(t)$ are commonly assumed independent with mean zero and variance one [9, 10, 11].

In continuous time, standard Brownian motion $W(t)$ starts at $W(0) = 0$, has continuous sample paths, has independent increments, and has stationary Gaussian increments: $W(t) - W(s) \sim N(0, t - s)$ for $t > s$. The increment variance is Δt , while its standard deviation is $(\Delta t)^{1/2}$. Although its increments are stationary, $W(t)$ itself is not: $\text{Var}(W(t)) = t$. White noise describes uncorrelated values at distinct dates; Brownian motion is a continuous-path process whose increments over disjoint intervals are independent [20].

Arithmetic Brownian motion adds drift and scale to that driver, $S(t) = S(0) + \mu t + \sigma W(t)$, so its changes can be positive or negative without keeping $S(t)$ positive. Geometric Brownian motion instead models proportional changes, $dS(t) = \mu S(t) dt + \sigma S(t) dW(t)$, giving $S(t) = S(0) \exp\{(\mu - \sigma^2/2)t + \sigma W(t)\}$ and positive prices for $S(0) > 0$ [14, 16, 17].

Fractional Brownian motion $B^H(t)$ replaces the Brownian scaling exponent $1/2$ with a Hurst parameter $H \in (0, 1)$. Its centered Gaussian increments are stationary and satisfy $\text{Var}(B^H(t + \Delta t) - B^H(t)) = (\Delta t)^{2H}$, so their standard deviation is $(\Delta t)^H$; $H = 1/2$ recovers standard Brownian motion. For $H > 1/2$, separated increments are positively correlated, and their slowly decaying covariance gives long memory. For $H < 1/2$, increments are negatively correlated and the paths are locally rougher. Negative correlation here does not imply long memory in the usual nonsummable-covariance sense [20, 19].

Ordinary continuity means a path's change tends to zero as the time interval tends to zero. Hölder continuity is stronger: locally, $|B^H(t) - B^H(s)| \leq C|t - s|^r$ for some finite C and any exponent $r < H$ on a bounded interval. A smaller H therefore permits less regular paths [19]. A power law has the form $C\Delta^{-\gamma}$: its log-log plot against lag Δ is a straight line with slope $-\gamma$. In *Volatility is rough*, Gatheral, Jaisson, and Rosenbaum find that volatility autocovariance does not follow such a power law at their usual observed time scales, even though standard persistence procedures can suggest long memory. They explicitly do not infer the covariance's behavior at unobservable asymptotically large lags from finite data [19].

To combine rough local behavior with mean reversion, fractional Ornstein–Uhlenbeck (fOU) log volatility $X(t)$ adds a restoring drift to fractional Brownian forcing: $dX(t) = -\alpha(X(t) - m)dt + \nu dB^H(t)$, with $\alpha > 0$. Its stationary form is $X(t) = m + \nu \int_{-\infty}^t e^{-\alpha(t-s)} dB^H(s)$: deviations from m decay at a rate set by α , while H controls local roughness. The rough fractional stochastic-

volatility model sets $\sigma(t) = \exp\{X(t)\}$ and takes $H < 1/2$. These are Equations (3.3)–(3.4) of Gatheral, Jaisson, and Rosenbaum; the fractional OU construction is also studied by Cheridito, Kawaguchi, and Maejima [19, 21].

2.2 ML Concepts

3 Methods

4 Implementation

5 Experimental Results

5.1 Goals

5.2 Setting

5.3 Case Studies

5.4 Correctness

5.5 Computational Evaluation

5.6 Technical Evaluation

6 Conclusions

References

- [1] Euronext. (2015). Gong marks exchange anniversary and exchange history website launch. Euronext Amsterdam.
- [2] Bloomberg Professional Services. (n.d.). Bloomberg Terminal: Real-time data, news, and analytics.
- [3] Nasdaq. (n.d.). Nasdaq TotalView: Direct market-data feed.
- [4] Zuckerman, G. (2019). *The man who solved the market: How Jim Simons launched the quant revolution*. Portfolio.
- [5] Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, 25(2), 383–417.

- [6] Yule, G. U. (1927). On a method of investigating periodicities in disturbed series, with special reference to Wolfer’s sunspot numbers. *Philosophical Transactions of the Royal Society of London. Series A*, 226, 267–298.
- [7] Slutsky, E. (1927). The summation of random causes as a source of cyclic processes. *Problems of Economic Conditions*, 3(1), 34–64. English translation in *Econometrica*, 5(2), 105–146 (1937).
- [8] Wold, H. (1938). *A study in the analysis of stationary time series*. Almqvist & Wiksell.
- [9] Box, G. E. P., and Jenkins, G. M. (1970). *Time series analysis: Forecasting and control*. Holden-Day.
- [10] Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.
- [11] Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.
- [12] Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196.
- [13] Müller, U. A., Dacorogna, M. M., Davé, R. D., Pictet, O. V., Olsen, R. B., and Ward, J. R. (1993). *Fractals and intrinsic time: A challenge to econometricians*. Olsen & Associates working paper.
- [14] Bachelier, L. (1900). Théorie de la spéculation. *Annales scientifiques de l’École Normale Supérieure*, 17, 21–86.
- [15] Osborne, M. F. M. (1959). Brownian motion in the stock market. *Operations Research*, 7(2), 145–173.
- [16] Samuelson, P. A. (1965). Rational theory of warrant pricing. *Industrial Management Review*, 6(2), 13–31.
- [17] Black, F., and Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3), 637–654.
- [18] Comte, F., and Renault, E. (1998). Long memory in continuous-time stochastic volatility models. *Mathematical Finance*, 8(4), 291–323.

- [19] Gatheral, J., Jaisson, T., and Rosenbaum, M. (2018). Volatility is rough. *Quantitative Finance*, 18(6), 933–949.
- [20] Mandelbrot, B. B., and Van Ness, J. W. (1968). Fractional Brownian motions, fractional noises and applications. *SIAM Review*, 10(4), 422–437.
- [21] Cheridito, P., Kawaguchi, H., and Maejima, M. (2003). Fractional Ornstein–Uhlenbeck processes. *Electronic Journal of Probability*, 8(3), 1–14.
- [22] Werbos, P. J. (1974). *Beyond regression: New tools for prediction and analysis in the behavioral sciences* [Doctoral dissertation, Harvard University].
- [23] Rumelhart, D. E., Hinton, G. E., and Williams, R. J. (1986). Learning representations by back-propagating errors. *Nature*, 323, 533–536.
- [24] Lapedes, A., and Farber, R. (1987). *Nonlinear signal processing using neural networks: Prediction and system modelling* (Technical Report LA-UR-87-1662). Los Alamos National Laboratory.
- [25] Waibel, A., Hanazawa, T., Hinton, G., Shikano, K., and Lang, K. J. (1989). Phoneme recognition using time-delay neural networks. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 37(3), 328–339.
- [26] Bai, S., Kolter, J. Z., and Koltun, V. (2018). An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. arXiv:1803.01271.
- [27] Reisenhofer, R., Bayer, X., and Hautsch, N. (2022). HARNet: A convolutional neural network for realized volatility forecasting. arXiv:2205.07719.
- [28] Elman, J. L. (1990). Finding structure in time. *Cognitive Science*, 14(2), 179–211.
- [29] Hochreiter, S., and Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735–1780.
- [30] Cho, K., van Merriënboer, B., Gülçehre, C., Bahdanau, D., Bougares, F., Schwenk, H., and Bengio, Y. (2014). Learning phrase representations using RNN encoder–decoder for statistical machine translation. *Proceedings of EMNLP*, 1724–1734.

- [31] Breiman, L. (2001). Random forests. *Machine Learning*, 45, 5–32.
- [32] Chen, T., and Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD Conference*, 785–794.
- [33] Vaswani, A., et al. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30.
- [34] Andersen, T. G., Bollerslev, T., Diebold, F. X., and Labys, P. (2003). Modeling and forecasting realized volatility. *Econometrica*, 71(2), 579–625.
- [35] Parkinson, M. (1980). The extreme value method for estimating the variance of the rate of return. *The Journal of Business*, 53(1), 61–65.
- [36] Garman, M. B., and Klass, M. J. (1980). On the estimation of security price volatilities from historical data. *The Journal of Business*, 53(1), 67–78.
- [37] Audrino, F., and Chassot, J. (2024). HARd to beat: The overlooked impact of rolling windows in the era of machine learning. arXiv:2406.08041.
- [38] Montero-Manso, P., and Hyndman, R. J. (2021). Principles and algorithms for forecasting groups of time series: Locality and globality. *International Journal of Forecasting*, 37(4), 1632–1653.
- [39] Liu, C., Tran, M.-N., Wang, C., Gerlach, R., and Kohn, R. (2023). Global neural networks and the data scaling effect in financial time series forecasting. arXiv:2309.02072.
- [40] Zhang, C., Zhang, Y., Cucuringu, M., and Qian, Z. (2022). Volatility forecasting with machine learning and intraday commonality. arXiv:2202.08962.
- [41] Mouti, S. (2023). Rough volatility: Evidence from range volatility estimators. arXiv:2312.01426.
- [42] Tang, S. H., Rosenbaum, M., and Zhou, C. (2023). Forecasting volatility with machine learning and rough volatility: Example from the crypto-winter. arXiv:2311.04727.
- [43] Brini, A. (2026). Forecasting realized volatility with time series foundation models: A comparison with econometric benchmarks. arXiv:2607.05291.